TESTING NEURAL NETWORKS ON MOBILE DEVICES

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ZADÁNÍ BAKALÁŘSKÉ PRÁCE

I. OSOBNÍ A STUDIJNÍ ÚDAJE

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III. PŘEVZETÍ ZADÁNÍ

Student bere na vědomí, že je povinen vypracovat bakalářskou práci samostatně, bez cizí pomoci, s výjimkou poskytnutých konzultací. Seznam použité literatury, jiných pramenů a jmen konzultantů je třeba uvést v bakalářské práci.

Datum převzetí zadání

Nguyen Hung Manh

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Pokyny pro vypracování:

The goal of this thesis is to experimentally evaluate the feasibility of running modern neural networks for face processing on mobile devices (Android, iOS). The student will compare different architectures, frameworks, and device types in terms of speed, energy consumption, and accuracy. The work will focus primarily on real-time face detection and facial landmark estimation from video.
Tasks
1. Familiarize yourself with the options for running neural networks on mobile platforms.
2. Select suitable models for facial landmark detection and collect several test videos.
3. Implement a testing scenario on both Android and iOS.
4. Evaluate latency (ms/frame), throughput (fps), and optionally energy consumption. Compare results across several different devices.
5. Optionally, explore various optimization methods (e.g., quantization) and evaluate their impact on speed and accuracy.
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Seznam doporučené literatury:

I would like to thank my supervisor Ing. Jan Čech, Ph.D. for his guidance, feedback, and support throughout this thesis.
I also thank my family and friends for their encouragement.

Declaration

I hereby declare that the presented thesis is my own work and that I have cited all sources of information in accordance with the Guideline for adhering to ethical principles when elaborating an academic final thesis.

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In Praze on May 5, 2026

Abstract

This thesis focuses on benchmarking for neural networks for detecting facial landmarks on Android and iOS devices. It discusses mobile hardware constraints, efficient model design, quantization principles, and the ExecuTorch runtime for mobile inference. In this thesis, a backend means the software execution path that runs supported model operations using a specific implementation, such as optimized CPU kernels or an Apple acceleration framework.

Keywords mobile inference, face detection, facial landmarks, benchmarking, quantization, ExecuTorch, Core ML, XNNPACK

Abstrakt

Práce se věnuje benchmarkingu neuronových sítí pro detekci obličejových landmarků na mobilních zařízeních (Android, iOS). Popisuje omezení mobilního hardwaru, principy efektivních architektur a kvantizace a runtime Executorch a proces inference na mobilních zařízeních.

Klíčová slova mobilní inference, detekce obličeje, obličejové body, benchmark, kvantizace, ExecuTorch, Core ML, XNNPACK

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List of abbreviations

AFW	Annotated Faces in the Wild (dataset)
AI	Artificial intelligence
ANE	Apple Neural Engine
AOT	Ahead-of-time
API	Application programming interface
AUC	Area Under the Curve
BS	Batch size
CNN	Convolutional Neural Network
CPU	Central Processing Unit
Core ML	Apple on-device machine learning framework
CSV	Comma-separated values
E2E	End-to-end
FLOPs	Floating-point operations
FP32	32-bit floating point
FPS	Frames per second
GPU	Graphics Processing Unit
INT8	8-bit integer quantization
IoU	Intersection over Union
JNI	Java Native Interface
LM	Landmark
NME	Normalized Mean Error
NPU	Neural Processing Unit
PC	Personal computer
PFLD	Practical Facial Landmark Detector
PIPNet	Pixel-in-Pixel Network
PTQ	Post-training quantization
PT2E	PyTorch 2 Export quantization flow
QAT	Quantization-aware training
ROI	Region of interest
SoC	System-on-Chip
XNNPACK	Optimized CPU execution path used by ExecuTorch

Chapter 1

Introduction

This thesis addresses problem of deploying and evaluating modern neural networks designed for detecting facial landmarks directly on mobile devices. The work focuses on the theoretical foundations and experimental methodology required for benchmarking of face detection and facial landmark estimation on Android and iOS using ExecuTorch and backend delegates.

1.1 Motivation

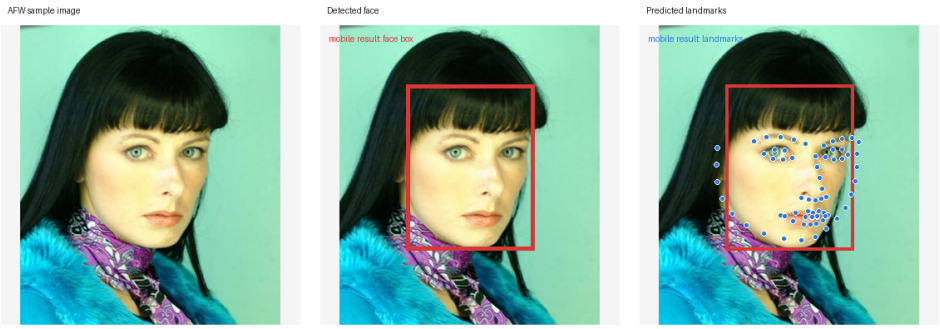
On-device inference has become a key requirement for many applications, because it improves privacy, reduces round-trip latency, and allows offloading compute resource to edge devices. At the same time, mobile devices are constrained by strict limits on compute throughput, memory bandwidth, power, and thermal budget [1, 2]. In practice, this means that achieving good model accuracy is not sufficient; the full end-to-end system must also be efficient.

Facial landmarks are useful because they describe the geometry of a face, not only whether a face is present. Landmarks mark points such as eye corners, nose position, mouth corners, and jaw shape. These points can be used to align faces before face recognition or verification [3], estimate head pose and localize faces in difficult real-world images [4], analyze facial expression, drive avatar or face-animation systems, and place augmented-reality effects such as glasses, masks, or makeup on the correct parts of the face [5]. In medical settings, landmark-based facial analysis can also help quantify facial asymmetry, support facial-palsy assessment, and track recovery or rehabilitation progress from facial movement changes [6].

Recent community benchmarks confirm that measured latency strongly depends on hardware configuration, runtime backend, and numerical precision [7, 2].

Figure **??** shows the practical meaning of the two-stage task before any performance numbers are discussed. The first panel is the image as it enters the application. The second panel shows face detection: the system answers the

Figure 1.1 Face-processing pipeline on a real AFW sample image: the first panel shows the input crop, the second panel draws the face-detection bounding box from a benchmark result, and the third panel overlays the predicted landmarks from the same result.



question "where is the face?" by drawing a rectangle around the face region. In this figure, the rectangle is taken from an exported mobile benchmark image record for the AFW sample. This rectangle is then used to crop and normalize the face before landmark estimation. The third panel shows the landmarks predicted by the same benchmark record: instead of only locating the face, the model predicts semantically meaningful points such as eyes, nose, mouth, and face contour positions. These points are what make downstream tasks such as face alignment, expression analysis, augmented-reality overlays, and pose estimation possible.

1.2 Problem statement

The core of this thesis is to evaluate neural networks for detecting facial landmarks on mobile devices, with a focus on configurations implemented with ExecuTorch. The benchmark must support equivalent iOS and Android applications, consistent measurement conditions, and validation of quality metrics.

The benchmark is a two-stage pipeline:

1. face detection in an input image,
2. facial landmark estimation on detected face crops.

1.3 Research objectives

Objectives of the thesis are:

- summarize practical options for running neural networks on Android and iOS;

- select suitable face-processing models (detector + landmark regressor) for mobile real-time use;
- implement equivalent benchmark scenarios on both platforms;
- evaluate latency (ms/frame), throughput (frames per second, FPS), and landmark accuracy;
- analyze post-training quantization (PTQ) with 8-bit integer (INT8) optimization impact on speed and quality.

1.4 Alignment with thesis assignment tasks

The assignment tasks are covered as follows:

1. **Mobile neural network (NN) deployment options:** Chapter 2 and Chapter 3 review mobile hardware/runtime constraints and ExecuTorch backend options.
2. **Model selection + test inputs:** Chapter 2 (model section) explains selected detector/landmark architectures; Chapter 4 defines the reproducible AFW (Annotated Faces in the Wild)-based input protocol used for quantitative comparison.
3. **Android/iOS testing scenario:** Chapter 4 documents feature parity and unified measurement schema for both mobile apps.
4. **Latency/FPS and optional energy:** Chapter 4 defines metrics and logging; Chapter 5 reports results measured on various devices.
5. **Optimization impact (quantization):** Chapter 3 explains PTQ export workflow and Chapter 5 compares 32-bit floating point (FP32) vs INT8 behavior.

Chapter 2

Theoretical background

2.1 On-device inference in mobile environments

Mobile artificial intelligence (AI) systems are typically deployed on heterogeneous Systems-on-Chip (SoCs), where central processing units (CPUs), graphics processing units (GPUs), and dedicated neural processing units (NPUs) coexist [8, 1]. Compared to server inference, mobile workloads operate under tighter thermal and power limits and often share resources with other tasks.

Latency depends on several things:

- model architecture (operator types, tensor shapes, parameter count),
- runtime backend (kernel library, operator fusion, threading policy),
- hardware mapping (CPU, GPU, and NPU dispatch),

That means the same neural network can show very different performance across devices and software stacks [2, 7].

2.2 Hardware and system constraints

2.2.1 CPU, GPU, and NPU roles

CPUs provide broad operator coverage and predictable behavior, but may not offer the highest throughput for dense tensor computations. Mobile GPUs can improve throughput for selected workloads, while NPUs may deliver superior energy-efficiency when supported operators map well to accelerator kernels [8, 1]. In real applications, unsupported subgraphs frequently fall back to CPU, and thus synchronization and memory transfer overhead become issues.

The difference between these three hardware types is important for interpreting mobile benchmark results. A CPU is the most general processor in the

phone. It is good at control flow, irregular work, small operations, and operators that are not supported by accelerators. This makes it reliable, but not always the fastest choice for large neural-network tensor operations. In this thesis, XNNPACK (optimized CPU backend) represents the optimized CPU path, so XNNPACK results should be understood as optimized CPU execution rather than GPU or NPU execution [9, 10].

A GPU is designed for many similar operations in parallel. Neural-network layers such as convolutions and matrix multiplications can often be expressed as large parallel workloads, so a GPU can be faster than a CPU for supported operators. The drawback is that GPU execution has overhead: data may need to be prepared for the GPU, command queues must be scheduled, and unsupported operators may have to move back to the CPU. For small models or small batch sizes, this overhead can reduce the practical speedup [8, 2].

An NPU is a dedicated accelerator for neural-network operations. It is usually designed to run supported layers with high energy efficiency. The limitation is operator coverage: if the model contains an unsupported layer, unsupported shape, or unsupported data type, the runtime may fall back to CPU execution for that part. This fallback can introduce extra copies and synchronization, so an accelerator is useful only when a large enough part of the graph maps cleanly to it [1, 7].

In this thesis, the measured backends are Core ML (Apple machine learning framework), XNNPACK, and portable ExecuTorch execution. Core ML can choose Apple CPU, GPU, or Apple Neural Engine (ANE) depending on the model and device [11, 12]. XNNPACK is the optimized CPU path [9]. Portable execution is the generic ExecuTorch path used mainly as a correctness baseline when no backend-specific acceleration is used [13, 14].

2.2.2 Power and thermal behavior

Peak performance is often not sustainable in mobile devices. Very long inference can trigger thermal throttling, reducing frequency and increasing latency [2].

2.3 Efficient neural networks for mobile vision

The shift from convolutional neural networks (CNNs) deployed on servers to mobile-friendly architectures was enabled by operator-level efficiency ideas such as depthwise separable convolution, inverted residual blocks, and channel shuffle [15, 16, 17]. SqueezeNet demonstrated that competitive accuracy can be achieved with significantly fewer parameters [18].

For processing faces on mobile devices, both detector and landmark models must be efficient.

2.4 Model compression and quantization

Compression and quantization reduce memory footprint and often improve runtime performance. The most common compression techniques include pruning and weight sharing [19]. In practical mobile deployment, integer quantization is especially important because INT8 kernels can be faster and more energy-efficient than FP32 on supported hardware [20].

2.4.1 Post-training quantization

Post-training quantization converts an already trained FP32 model to a lower-precision representation, usually without retraining. In this thesis, the main target is INT8 execution. A real value x is represented by an integer q using scale s and zero-point z:

$$q = \mathrm{clip}\Big(\mathrm{round}\Big(\frac{x}{s}\Big) + z,\ q_{\min}, q_{\max}\Big), \qquad \hat{x} = s\,(q - z). \tag{2.1}$$

For signed INT8, q is typically in $[-128, 127]$ (or $[-127, 127]$ in symmetric weight quantization). The reconstructed value $\hat{x}$ approximates x, so a small quantization error is introduced.

The scale s is the size of one integer step in the original real-number space. If s is small, the integer grid is fine and precise, but the representable range is narrow. If s is large, the range is wide, but nearby real values are rounded to the same integer. The zero-point z says which integer represents real value zero. In symmetric quantization, z is normally zero or very close to zero, which makes signed integer arithmetic simpler [20]. In asymmetric quantization, z can shift the integer range so that non-symmetric activation ranges can be represented more efficiently [21].

Calibration is the step that chooses useful ranges for tensors. During calibration, representative input samples are passed through the prepared model, and observer nodes record approximate minimum and maximum values of activations [22]. These observed ranges are then converted into scales and zero-points. If calibration data is too narrow, later real inputs may fall outside the observed range and get clipped by the clip operation in the equation above. If calibration data is too wide because of rare outliers, the scale becomes too large and normal values lose precision. This is why PTQ must be evaluated experimentally instead of being assumed to preserve accuracy.

For weights, per-tensor quantization uses one scale for the whole tensor. Per-channel quantization uses a separate scale for each output channel, so one channel with large values does not force all other channels to use a coarse scale [20]. For convolution and linear layers, this is often important because different output channels can have very different value ranges.

PTQ typically has three steps:

1. prepare the graph for quantization,

2. run calibration data through the model to estimate activation ranges,
3. convert operators to quantized kernels.

Compared to quantization-aware training (QAT), PTQ is much simpler to apply, but it is more sensitive to calibration quality and to model-specific outliers.

2.4.2 INT8 design choices that affect quality

Two design choices are central for this work:

- **Per-tensor vs per-channel weights:** per-channel assigns separate scales per output channel and usually preserves accuracy better, especially in convolution-heavy models.
- **Static vs dynamic activations:** static PTQ calibrates activation ranges offline; dynamic approaches estimate part of the range online. Static PTQ is often faster when backend support is good.

INT8 often improves speed for two reasons: lower memory traffic (INT8 tensors are 4× smaller than FP32 tensors) and faster vectorized kernels. However, real inference latency reduction depends on operator coverage. If part of the graph falls back to non-quantized execution, conversion overhead can reduce the expected speedup [21].

2.5 Face detection and landmark estimation

2.5.1 Task definitions

Face detection finds faces with bounding boxes. Landmark estimation predicts keypoint coordinates (e.g., eye corners, nose tip, mouth corners), usually on a cropped face image. These tasks are strongly coupled: detection quality and crop geometry directly influence landmark quality.

2.5.2 Representative models

RetinaFace is a strong one-stage detector with high robustness under challenging conditions [23, 24]. FaceBoxes is a lightweight detector explicitly designed for CPU real-time operation [25, 26].

For landmarks, PFLD (Practical Facial Landmark Detector) emphasizes compactness and speed [5, 27]. MobileFaceNet provides a lightweight backbone originally developed for mobile face recognition and often reused for landmark regression [3, 28]. PIPNet (Pixel-in-Pixel Network) combines heatmap-style reasoning with low-resolution efficient heads for practical landmark localization [29, 30].

In the benchmark matrix used in this thesis, the main detector+landmark combinations are based on RetinaFace-MNet0.25 or FaceBoxesV2 with MobileFaceNet, PFLD-MNv2-0.25, PFLD-MNv2-1.0, and PipNet-ResNet18.

Table 2.1 Qualitative properties of models used in the benchmark matrix.

Model	Stage	Relative cost	Main properties
RetinaFace-MNet0.25	Detector	Medium	Robust detection quality; higher detector cost than very light alternatives.
FaceBoxesV2	Detector	Low	Very fast detector; useful when throughput is prioritized.
MobileFaceNet	Landmark	Low	Lightweight baseline; stable and efficient for mobile CPUs.
PFLD-MNv2-0.25	Landmark	Low	Compact model with good speed and moderate landmark precision.
PFLD-MNv2-1.0	Landmark	Medium	Higher capacity than 0.25 variant; better potential quality at higher compute cost.
PipNet-ResNet18	Landmark	High	Stronger landmark localization capacity, but substantially heavier compute load.

2.5.3 Datasets and annotation uncertainty

In this thesis, AFW annotations are used for wild-condition landmark evaluation, originating from the face detection and pose estimation work of Zhu and Ramanan [4].

Chapter 3

ExecuTorch for mobile deployment

3.1 ExecuTorch motivation

ExecuTorch is a PyTorch edge deployment stack designed for efficient on-device inference across mobile and embedded devices [13, 14]. Its central design principle is *ahead-of-time* (AOT) preparation: expensive transformation and planning steps are performed before deployment, so runtime execution remains lightweight.

ExecuTorch was chosen instead of TensorFlow Lite (TFLite) or another mobile runtime mainly because this project starts from PyTorch models and needs the same deployment idea on Android and iOS. TFLite is a strong mobile and embedded runtime for models in the TensorFlow Lite format [31], and ONNX Runtime Mobile is a strong mobile runtime for models represented in the Open Neural Network Exchange (ONNX) format [32]. However, using either of them here would add an extra conversion boundary: the model would first have to leave the PyTorch export path and then be validated again in another framework-specific graph format. That extra step is risky for this benchmark because small differences in operator conversion, tensor layout, unsupported operations, or numerical precision could change both speed and landmark accuracy.

ExecuTorch keeps the pipeline closer to PyTorch: `torch.export` captures the model graph, ExecuTorch lowers that graph for edge execution, and the final `.pte` file is loaded by the Android and iOS benchmark applications [33, 34, 35]. It also matches the backend comparison needed in this thesis. The same deployment stack can produce an optimized CPU path through XNNPACK, an Apple path through Core ML, and a portable ExecuTorch baseline [9, 12, 13]. This makes the benchmark easier to interpret: differences in the results are more directly connected to backend choice, quantization, device hardware,

and model architecture instead of being mixed with a completely different model-conversion toolchain.

3.2 Conceptual architecture

ExecuTorch organizes model deployment into three phases [13, 14]:

1. export a PyTorch model to graph form,
2. lower and optimize the graph for edge execution,
3. execute the serialized program with a small C++ runtime.

The first phase, export, captures the model as a graph. A graph is a list of tensor operations and their data dependencies. This graph form is important because it can be checked, transformed, quantized, partitioned, and serialized [33, 34].

In more detail, the graph is a data-flow description of the neural network. Each node represents something that must happen, such as reading an input tensor, reading a trained weight tensor, running a convolution, applying an activation function, reshaping a tensor, or returning the final output. The edges between nodes represent values flowing from one operation to another [33]. If node B uses the result of node A, then the graph records that dependency. This lets the compiler know that A must run before B, and it also lets the compiler see when the output of A is no longer needed.

The graph is an internal program representation. It contains placeholders for model inputs, constants or parameters for learned weights, operator calls, metadata about tensor shapes, and output nodes [33, 13]. A tensor is the multi-dimensional array used by neural networks. For example, an image batch may be represented as a tensor with dimensions for batch, height, width, and channels. Operator nodes consume tensors and produce new tensors. A convolution node consumes an input feature tensor and a weight tensor, then produces an output feature tensor. The next activation node consumes that output and produces another tensor.

This representation is useful because it makes implicit model behavior explicit. In normal PyTorch eager execution, Python code decides what operation to run next while the model is executing. In an exported ExecuTorch workflow, that decision has already been captured into the graph [13]. The phone therefore does not need to interpret the original Python model. It only follows the prepared graph program.

Technically, `torch.export` captures an Export Intermediate Representation, often called Export IR, that preserves the meaning of the original PyTorch program while removing the need to run the original Python control code on the phone [33, 13]. The exported graph records operators, tensor shapes or

shape constraints, parameters, and the order in which values flow between operators [33]. This is why ExecuTorch can analyze the model before deployment instead of discovering the model structure during each mobile inference run.

Shape information is part of why the graph can be compiled. If an input always has shape $1 \times 3 \times 112 \times 112$, the compiler can plan memory for that exact shape. If the batch dimension is dynamic, the graph can store a constraint such as "batch is between 1 and 128" instead of a single fixed number [33]. ExecuTorch can then use these constraints during lowering, memory planning, and backend partitioning.

After export, compiler passes can rewrite the graph. A pass is a transformation that reads the graph and produces a modified graph or program. Typical passes can decompose high-level PyTorch operators into simpler Core ATen operators, insert or remove quantization-related operations, identify which nodes can be handled by XNNPACK or Core ML, and assign temporary tensors to planned memory locations [13, 14, 36, 37]. Because the model is represented as nodes and edges, these transformations can be done systematically rather than by manually editing model code.

The second phase, compile or lower, turns the exported graph into an ExecuTorch program. During this phase ExecuTorch can decompose operators into a smaller standardized operator set, lower supported graph regions to backends such as XNNPACK or Core ML, and plan temporary tensor memory [14, 36, 37].

ExecuTorch uses operator standardization to make this possible. Many PyTorch operators are decomposed into a smaller Core ATen operator set, where ATen is PyTorch's core tensor and operator library, and the portable operator library implements this standardized set when no delegate handles the operation [13, 14]. This matters because backend authors do not need to support the entire Python-level PyTorch API; they only need to support the graph operators that the exported model contains.

The third phase, execution, is intentionally small. The runtime reads the serialized program, prepares memory, calls either portable kernels or delegate backends, and returns output values. The runtime is written in C++ and is wrapped by platform application programming interfaces (APIs) for Swift and Kotlin/Java [13, 38, 39].

The runtime is also designed so that only the operators needed by the deployed program have to be linked into the final app binary [13]. In simple terms, the phone does not ship a full Python interpreter or a full training framework. It ships a small execution engine, the required kernels, and any registered delegate backends needed by the exported model.

3.3 Export and lowering workflow

The practical workflow consists of:

1. model export using `torch.export`,
2. backend specific lowering via `to_edge_transform_and_lower`,
3. creation of a serialized `.pte` artifact, where an artifact means a file produced by the build/export pipeline and used later by another part of the project.

In this thesis, a model artifact is usually a `.pte` file: it is produced on the development machine, copied into the Android or iOS application, and loaded by the mobile benchmark.

This process supports dynamic shape constraints and optional separation of program and tensor data into `.pte` and `.ptd` files [34, 35, 40, 33].

3.3.1 Intermediate representations used before runtime

ExecuTorch does not directly run the original Python model on the phone. The model passes through intermediate representations, meaning machine-readable forms of the model that are easier for compiler passes to analyze and modify [14]. The important idea is:

- **PyTorch model**: the normal Python model used during development;
- **Exported graph**: a graph produced by `torch.export`, with tensor operations and shape constraints [33];
- **Edge program**: an ExecuTorch-friendly program after edge transformations, backend lowering, and memory planning [34, 37];
- **Serialized artifact**: the final `.pte` (portable tensor executable) file copied to Android assets or the iOS bundle [35].

This is why the benchmark applications do not contain PyTorch model code. They only load already exported artifacts and feed tensors into them. In simple language, Python is used to prepare the model, and the phone only receives the finished program.

3.3.2 The `.pte` deployment artifact

The `.pte` file is the serialized ExecuTorch program used at runtime [35]. It contains the graph-level execution plan, operator references, delegate information, constants, and metadata needed by the ExecuTorch runtime. If large tensor data is separated from the program, a `.ptd` file can store that tensor data separately [40]. This thesis mainly treats `.pte` files as the deployable model artifacts: the Android and iOS applications load them, create input tensors, run forward execution, and read output tensors.

A `.pte` file is therefore not the same thing as a PyTorch checkpoint. A checkpoint stores trained parameters, and often assumes that Python model code will rebuild the network structure. A `.pte` file stores an already exported and lowered execution program. That program can include delegate segments, portable fallback segments, constants, and the method entry points that the runtime can call [35, 36].

3.3.3 Static shapes, dynamic shapes, and batching

Mobile runtimes are easier to optimize when tensor shapes are known in advance. A static-shape export fixes dimensions such as batch size and input height and width before deployment. A dynamic-shape export allows selected dimensions to vary inside declared bounds using `torch.export.Dim` constraints [33]. In this thesis, dynamic batch export means that the batch dimension can vary between a minimum and maximum value while image shape and channel count remain fixed. This allows one artifact to cover several batch sizes, but it also gives the compiler less exact shape information than a fully static export.

3.3.4 Memory planning

ExecuTorch performs memory planning during export and lowering so that the runtime can reuse buffers for temporary tensors [37]. This matters on mobile devices because memory allocation during repeated inference can add overhead and increase memory pressure. A simple way to think about it is that many intermediate tensors are not needed at the same time. If tensor A is no longer used before tensor B is created, both can reuse the same memory region. Planning this before runtime helps make execution more predictable.

3.4 Delegation and partitioning

ExecuTorch uses delegates to offload supported subgraphs to backend specific implementations [36]. A partitioner tags lowerable graph regions; unsupported regions remain in portable execution. This mixed execution model improves portability but can create overhead.

For this thesis, two backends are central:

- **XNNPACK**: optimized CPU backend with broad mobile support, including quantized paths [9, 41, 10];
- **Core ML delegate**: Apple backend path that can dispatch to CPU, GPU, or ANE depending on model and platform capabilities [12, 11].

A partitioner is the component that decides which graph regions a delegate can own [36]. If a sequence of operators is supported by XNNPACK, that

region is marked for XNNPACK. If one operator in the middle is not supported, ExecuTorch may split the graph into multiple delegated regions with a portable fallback region between them [36]. This is important for the results chapter because the fastest model is not always the model with the fewest theoretical floating-point operations (FLOPs); backend coverage and fallback boundaries also matter.

Delegation is not just a runtime switch. It is decided during export and lowering. The partitioner examines the graph, checks backend support, and replaces supported graph regions with calls into a backend-specific lowered representation [36]. At runtime, ExecuTorch follows the already prepared execution plan. This is why two `.pte` files from the same neural network can behave differently: one may contain XNNPACK-delegated regions, one may contain Core ML-delegated regions, and one may contain only portable execution.

3.4.1 XNNPACK

XNNPACK is the CPU acceleration path used on both Android and iOS. It provides optimized neural network operators for mobile and desktop processors [10]. In ExecuTorch, the XNNPACK backend lowers supported graph regions into XNNPACK delegate calls [9]. This is useful for this thesis because it provides a common backend across Android and iOS, so the benchmark can compare the same exported family of artifacts on both platforms.

XNNPACK is a low-level library of optimized neural-network operator kernels, not a full training framework [10]. It targets CPU instruction sets on Arm and x86 systems, including Android, iOS, macOS, Linux, Windows, and simulators [10, 9]. The operators relevant to vision models include convolution, depthwise convolution, pooling, resize, fully connected layers, elementwise arithmetic, activation functions such as clamp/ReLU, sigmoid, tanh, and PReLU, and layout-related operations such as transpose [10]. This operator coverage is why XNNPACK is a reasonable backend for face detectors and landmark regressors, which are mostly made from convolution, activation, pooling, and linear layers.

The important technical point is that XNNPACK accelerates CPU execution by using specialized kernels and vectorized instructions. It does not move the model to the GPU or NPU. XNNPACK also commonly uses NHWC tensor layout, where the tensor dimensions are ordered as batch, height, width, and channels [10]. If an exported graph already matches what XNNPACK supports, the partitioner can delegate large regions. If the graph contains an unsupported operator or unsupported shape pattern, ExecuTorch must keep that part in portable execution, which can add boundary overhead.

For quantized models, the ExecuTorch XNNPACK backend supports 8-bit symmetric weights with 8-bit asymmetric activations through the PT2E flow [41]. The official XNNPACK quantization flow uses `XNNPACKQuantizer`,

`prepare_pt2e`, calibration with representative inputs, `convert_pt2e`, and then lowering with `XnnpackPartitioner` [41]. This matches the export script used in this thesis.

3.4.2 Core ML

Core ML is the Apple-specific delegate path [11]. In ExecuTorch, the Core ML backend lowers supported model regions to Apple's machine learning stack [12]. Depending on the model and device, Core ML may execute work on the CPU, GPU, or ANE [11]. This means that Core ML numbers in the results chapter should not be interpreted as CPU-only numbers.

In the ExecuTorch export flow, Core ML is selected by using the Core ML partitioner during the edge lowering step [12]. The partitioner marks graph regions that can be compiled for Core ML, and the resulting `.pte` file contains the information needed to call the registered Core ML delegate at runtime [12, 36]. No separate model conversion is performed inside the benchmark app; the app loads the already prepared ExecuTorch program.

Core ML supports dynamic dispatch to Apple CPU, GPU, and ANE and supports FP32 and 16-bit floating point (FP16) computation through the ExecuTorch backend [12]. This is why Core ML can be much faster than portable execution on iPhone. The measured time includes the full benchmark pipeline, but the actual landmark model forward pass may run on Apple acceleration hardware rather than only on the CPU. Core ML delegated models also depend on platform support: the ExecuTorch Core ML backend documents iOS 16 or newer as a target requirement for running delegated models [12].

Core ML and XNNPACK therefore answer different questions in the thesis. XNNPACK answers: "How fast is the optimized CPU backend that is available on both Android and iOS?" Core ML answers: "How fast can the Apple-specific delegate run when it is allowed to use Apple's ML stack?" Portable execution answers: "What happens without backend-specific acceleration?"

3.5 Quantization in ExecuTorch

ExecuTorch implements quantization through backend-specific configurations. These commonly use PT2E (PyTorch 2 Export) with torchao-based preparation and conversion [42, 22]. In this project, the XNNPACK INT8 export script produces the quantized landmark models using symmetric PTQ.

3.6 Runtime integration in the benchmark applications

On iOS, the thesis app uses the Swift ExecuTorch API (application programming interface). The app creates a module from a model path, creates a tensor

with the expected shape, calls the module forward method, and reads the returned tensor values. This matches the official iOS integration pattern for loading and executing an ExecuTorch model [39].

On Android, the thesis app uses the Java/Kotlin ExecuTorch API (application programming interface). The app loads a module from an asset path, wraps input data in a tensor, passes the tensor through `EValue` (ExecuTorch value wrapper) objects, and calls `forward`. The Java API is a wrapper over native ExecuTorch code and uses JNI (Java Native Interface) to cross between Java/Kotlin and C++ runtime code [38].

The important practical consequence is that neither mobile application trains or exports a model. Training and export happen before deployment. The mobile applications only run inference, meaning they load the `.pte` (serialized ExecuTorch program), feed input tensors, measure time, and store outputs for later analysis.

Chapter 4

Benchmark methodology

4.1 Methodological goals

The benchmark design in this thesis follows two goals:

1. measure latency, throughput, and accuracy consistently on iOS and Android,
2. preserve raw artifacts needed for analysis of the results.

4.2 Evaluated pipeline

Each benchmark run executes an end-to-end pipeline:

1. detector preprocessing,
2. detector inference,
3. detector postprocessing (box decoding and filtering),
4. face crop generation,
5. landmark model preprocessing,
6. landmark inference,
7. landmark postprocessing.

The detector part is always executed with detector batch size one in the reported pipeline runs. This means one image is normalized, passed through the detector, and decoded at a time. The landmark part has a separate *resolved landmark batch size*; this is the BS (batch size) value shown in the result tables and charts.

For a measurement iteration, the app first runs detection for the input image(s) and stores all detected boxes. It then creates face crops from those boxes. Finally, the generated crops are appended to a landmark queue and processed in chunks of the configured landmark batch size. This is why the thesis reports detector, crop, landmark-inference, and end-to-end timings separately: they correspond to distinct parts of the measured pipeline.

The reported thesis comparisons use 10 warm-up iterations before measurement. Warm-up iterations execute the same pipeline but are excluded from the aggregated timing statistics.

4.3 Model families and configuration space

The methodology supports multiple detector/landmark pairs and backend configurations. Typical factors include:

- detector choice (e.g., RetinaFace vs FaceBoxes),
- landmark architecture (e.g., PFLD, MobileFaceNet, PIPNet),
- precision variant (FP32 and INT8),
- backend delegate (XNNPACK, Core ML, and portable),
- resolved batch size.

A complete run record stores these factors explicitly, so later analysis does not depend on file naming conventions alone.

4.4 iOS and Android implementation parity

Both iOS and Android benchmark applications are designed to be feature-equivalent at the pipeline level. Equivalent behavior includes:

- identical preprocessing parameters,
- identical postprocessing logic and thresholds,
- consistent metric collection and export schema,
- consistent warm-up and measured iterations.

4.5 Dataset and inputs

AFW-based landmark annotations are used for evaluation of the accuracy [4].

4.6 Metrics

4.6.1 Landmark inference latency

Although each run executes the full detector+landmark pipeline, the thesis comparison charts are organized by the *landmark* configuration (model, backend, quantization, and resolved batch size). The primary timing metric is therefore landmark *inference* latency only:

$$\ell_i^{\text{lm}} = t_i^{\text{end}} - t_i^{\text{start}}. \tag{4.1}$$

For each run, this metric is extracted from per-iteration landmark-inference samples and summarized in milliseconds by average, $p50$, and $p95$. For charting, values are aggregated per configuration group using the median across repeated runs:

$$\tilde{m}_g = \text{median}\{m_r \mid r \in g\}. \tag{4.2}$$

4.6.2 Throughput and batch scaling

The benchmark applications export `fps`. It is run-level benchmark throughput, not landmark-forward-pass-only FPS. The forward-pass behavior of the landmark network is reported separately by the landmark-inference latency metric defined above. This distinction matters because an optimized landmark model can have very low forward latency while the measured run FPS is still limited by detector execution, preprocessing, postprocessing, queue flushing, or other surrounding pipeline work.

Run throughput follows the exported timing summary:

$$\text{FPS}_{\text{run}} = \frac{N_{\text{images}}}{T_{\text{run}}}. \tag{4.3}$$

Here, FPS means frames per second, N_{images} is the number of processed images, and T_{run} is the measured benchmark-run wall-clock time.

Batch size one (BS=1) charts compare median run FPS by landmark configuration. Batch-scaling charts plot median run FPS as a function of resolved landmark batch size ($b \in \{1, 2, 4, 8\}$ in the curated thesis set).

4.6.3 Landmark accuracy

Accuracy is recomputed offline from exported per-image predicted landmarks and AFW ground truth annotations [4]. For N evaluated images and K landmarks, inter-ocular normalized mean error (NME) is

$$\text{NME}_{\text{iod}} = \frac{1}{N}\sum_{n=1}^{N}\frac{1}{K}\sum_{k=1}^{K}\frac{\|\hat{\mathbf{p}}_{n,k} - \mathbf{p}_{n,k}\|_2}{d_n^{\text{iod}}}, \tag{4.4}$$

where d_n^{iod} is the inter-ocular distance for image n.

4.6.4 Cumulative error histograms

The result chapter also uses cumulative error histograms, sometimes called cumulative distribution curves. They are built from per-image errors rather than from a single average. For each evaluated image, the offline script computes an error value such as NME or RMSE. The script then asks a simple question for many thresholds on the horizontal axis: "what fraction of images has error less than or equal to this threshold?" The vertical axis is that fraction.

This representation is useful because average error can hide failures. Two models may have the same mean NME, but one can be reliable on almost every image while the other is very accurate on easy images and very poor on difficult images. In a cumulative histogram, the better curve is generally higher and more to the left: it means more images reach low error thresholds. A steep curve means consistent behavior and a long right tail means that some images are outliers where the landmark prediction failed or became noticeably worse.

4.6.5 Optional energy metrics

If platform tooling allows reliable collection, energy-related signals can be logged as supplementary indicators. In this thesis, landmark-inference latency, throughput, and landmark accuracy remain primary. Energy metrics are very hard to obtain. They are not included in the main results chapter, but they are collected when possible and stored in the raw artifacts for future analysis.

4.7 Measurement protocol

To reduce noise and measurement bias, each benchmark condition follows a fixed protocol:

- dedicated warm-up iterations before timed measurement,
- fixed number of measured iterations,
- stable device state (charging policy, background app control, thermal preconditioning),
- separate logging for initialization, inference, and optional postprocessing stages.

4.8 Run artifacts and offline analysis

Every run exports a machine-readable summary plus raw samples. Offline analysis scripts recompute aggregate statistics and recompute accuracy from image predictions instead of trusting prefilled summary fields. This allows for

more flexible and transparent analysis, as well as the possibility to add new metrics without rerunning benchmarks.

4.9 Threats to validity

Main threats and mitigations include:

- **Thermal drift**: controlled warm-up and repeated runs;
- **Backend asymmetry**: shared-configuration comparisons for cross-device claims;
- **Unsupported operators**: explicit reporting of delegate coverage and fallback paths;

Chapter 5

Experimental results

5.1 Result scope and source artifacts

The main simplified dashboard is configured around one focus mobile device (Xiaomi Redmi Note 8 Pro) with personal computer (PC) reference rows. Additional figures and tables include iPhone 14 results so that the reader can compare all three execution environments. When a chart compares PC, iPhone 14, and Redmi directly, it uses the same XNNPACK INT8 landmark configuration at the same batch size. This keeps the comparison focused on device and runtime behaviour instead of mixing different model variants.

5.2 Batch size 1 speed and latency overview

Figures **??** and **??** quantify the gap between desktop and mobile execution at BS=1. Batch size one means that the landmark network receives one detected face crop at a time, which is the most natural setting for an interactive camera application where frames arrive one by one. The FPS chart should be read as a measured run-throughput indicator: if FPS is low, the application cannot update landmark predictions smoothly. The latency chart should be read as a model-level diagnostic: it shows whether the landmark network itself is expensive, even if the full pipeline also contains detection, cropping, and post-processing.

The PC reaches roughly 16–25 run FPS across the main landmark models, while the focus phone falls into a narrower 3–8 run-FPS band. The gap is not uniform. It is smallest for the lightweight PFLD-MNv2-0.25 (PC/phone $\approx 2.0\times$ in run FPS) and largest for PipNet-R18 in FP32, where the PC is roughly an order of magnitude faster (24.86 vs 2.28 run FPS, i.e. $\sim 10.9\times$). The practical implication is that a model that looks comfortable on a desktop can still be too slow on a phone unless the mobile backend, quantization, and end-to-end pipeline are measured directly.

Figure 5.1 BS=1 median run FPS: PC vs the focus Android device. Here FPS is measured benchmark-run throughput, not landmark-forward-pass-only FPS; higher bars are better.

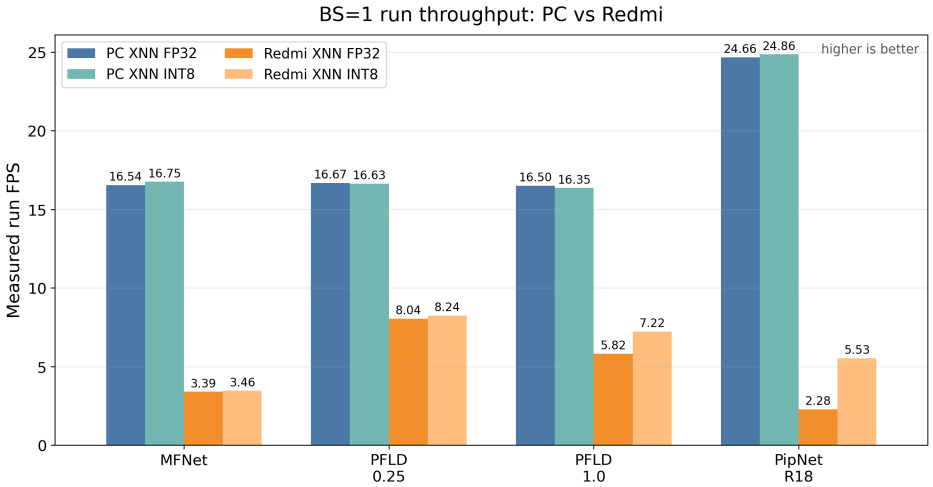
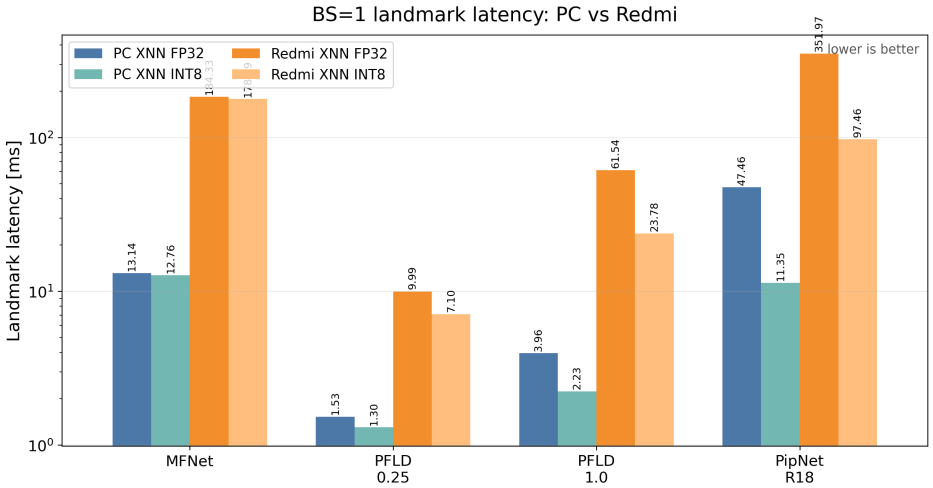
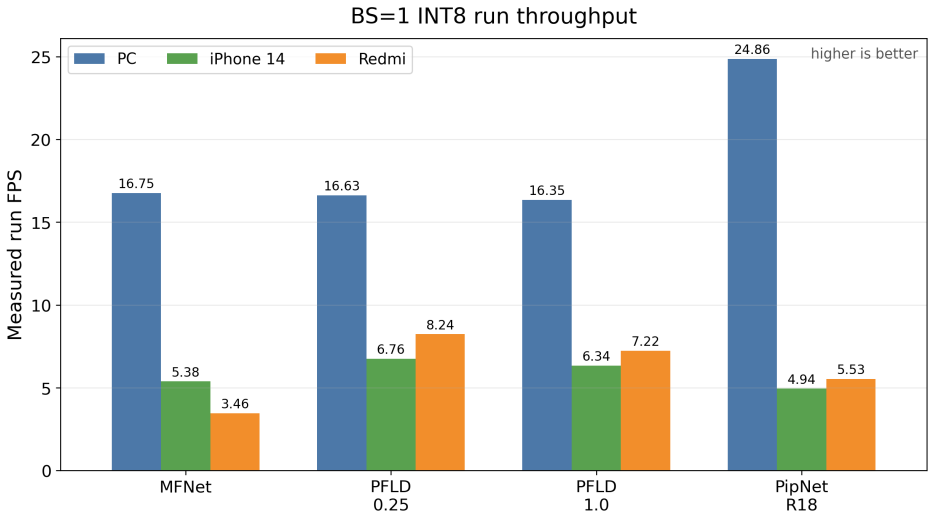


Figure 5.2 BS=1 median landmark-inference latency: PC vs the focus Android device. This measures only the landmark model forward pass in milliseconds, so lower bars are better.



5.3 Three-device BS=1 comparison

Figure 5.3 BS=1 INT8 XNNPACK run throughput on PC, iPhone 14, and Xiaomi Redmi Note 8 Pro. All bars use the same landmark backend and quantization, so higher run FPS means the measured benchmark run is faster on that device for the selected model.



Figures **??**, **??**, and **??** add the missing three-way comparison: PC, iPhone 14, and Redmi are shown in the same plots. The charts intentionally use XNNPACK INT8 for all three devices. This is not the fastest possible iPhone configuration, because the later iPhone-specific section shows that Core ML is usually better on Apple hardware. The purpose here is different: by holding the landmark backend and precision constant, the chart makes a fairer like-for-like comparison of the XNNPACK path.

The run-throughput chart shows the PC as the strongest reference platform, reaching about 16–25 run FPS depending on the model. The two phones are much closer to each other than either phone is to the PC. The iPhone 14 is faster than the Redmi for MobileFaceNet in this XNNPACK INT8 setting (5.38 vs 3.46 run FPS), while the Redmi is faster for PFLD-MNv2-0.25, PFLD-MNv2-1.0, and PipNet-R18 (8.24 vs 6.76 run FPS, 7.22 vs 6.34 run FPS, and 5.53 vs 4.94 run FPS). This does not mean that the Redmi is generally a faster device than the iPhone 14. It means that this particular CPU/XNNPACK benchmark path and this particular pipeline favour the Redmi on several models, while Apple-specific acceleration must be considered separately.

The end-to-end latency chart explains why run FPS alone is not enough. End-to-end latency includes all measured pipeline stages, so it answers the question a user would care about: “how long until this image has a landmark result?” The PC stays near 40–61 ms for the INT8 XNNPACK rows, while mobile end-to-end latency is typically above 100 ms. The landmark-only chart

Figure 5.4 BS=1 INT8 XNNPACK end-to-end latency on PC, iPhone 14, and Xiaomi Redmi Note 8 Pro. End-to-end latency includes detector execution, landmark execution, cropping, and surrounding pipeline work; lower is better.

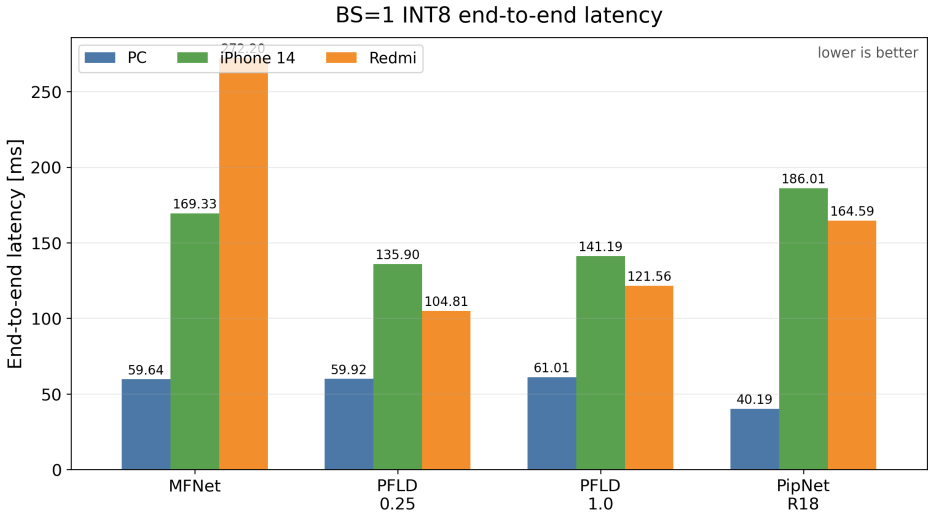
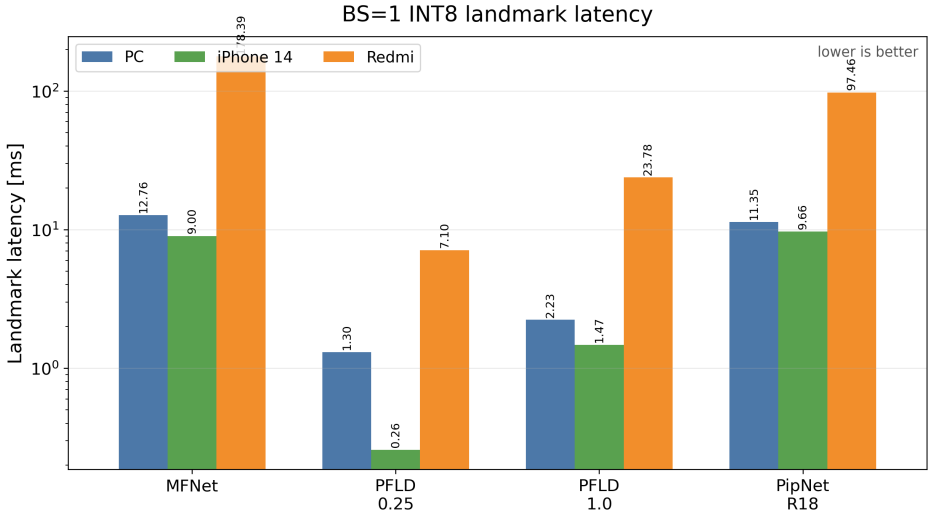


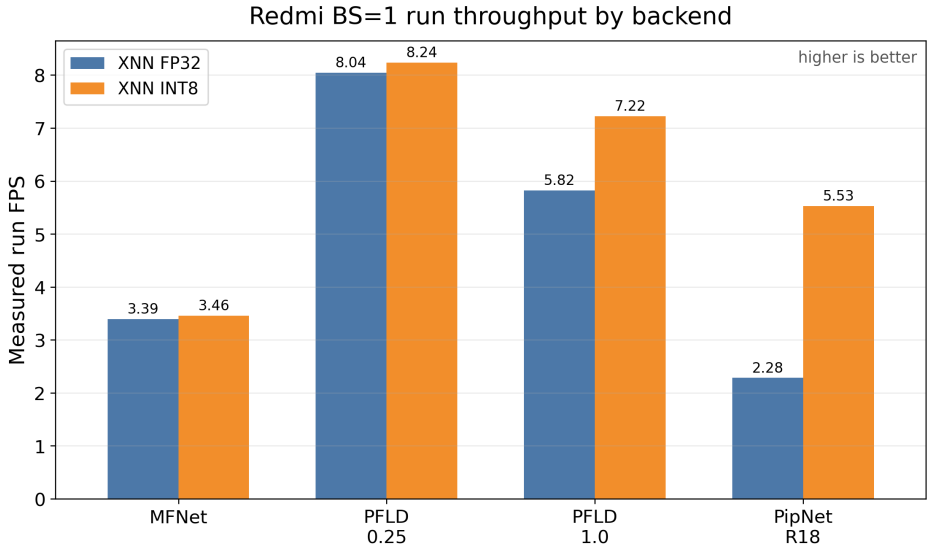
Figure 5.5 BS=1 INT8 XNNPACK landmark-inference latency on PC, iPhone 14, and Xiaomi Redmi Note 8 Pro. The vertical axis uses a logarithmic scale because the fastest and slowest landmark forward passes differ by orders of magnitude.



then separates the model forward pass from the rest of the pipeline. For PFLD-MNv2-0.25, iPhone 14 landmark inference is extremely small (0.26 ms), but end-to-end latency still remains around 136 ms. This shows that the detector and surrounding pipeline work can dominate once the landmark model is optimized. For PipNet-R18, the Redmi landmark forward pass is still expensive (97 ms), so quantization and model choice remain central to deployment feasibility.

5.4 Focus-device BS=1 backend and quantization

Figure 5.6 Focus device: BS=1 measured run throughput by backend and quantization.



PipNet-R18 gains 2.42× in run FPS (2.28 → 5.53) and its landmark inference drops from 352 ms to 97 ms (a 3.6× reduction); PFLD-MNv2-1.0 gains 1.24× in run FPS but 2.6× in landmark latency (61.5 → 23.8 ms).

The accuracy picture is correspondingly model-dependent. NME for MobileFaceNet is essentially unchanged (0.0428 to 0.0428, $< 0.1\%$ relative), and PipNet-R18 only degrades by $\sim$3.7% relative (0.0446 to 0.0463). The PFLD family is more sensitive. PFLD-MNv2-0.25 loses $\sim$6% relative NME (0.0978 to 0.1039), and the heavier PFLD-MNv2-1.0 loses $\sim$27% relative (0.0798 to 0.1014). The PFLD-MNv2-1.0 case is the canonical speed/accuracy trade-off in this benchmark: a 24% run-FPS gain costs a 27% NME increase, which may or may not be acceptable depending on the actual use case. PipNet-R18 INT8, by contrast, is a near-Pareto win. It has the largest speed gain in the matrix with only a barely measurable accuracy loss, making it the strongest INT8 candidate among the tested models on this device.

Figure 5.7 Focus device: BS=1 NME accuracy by backend and quantization (lower is better).

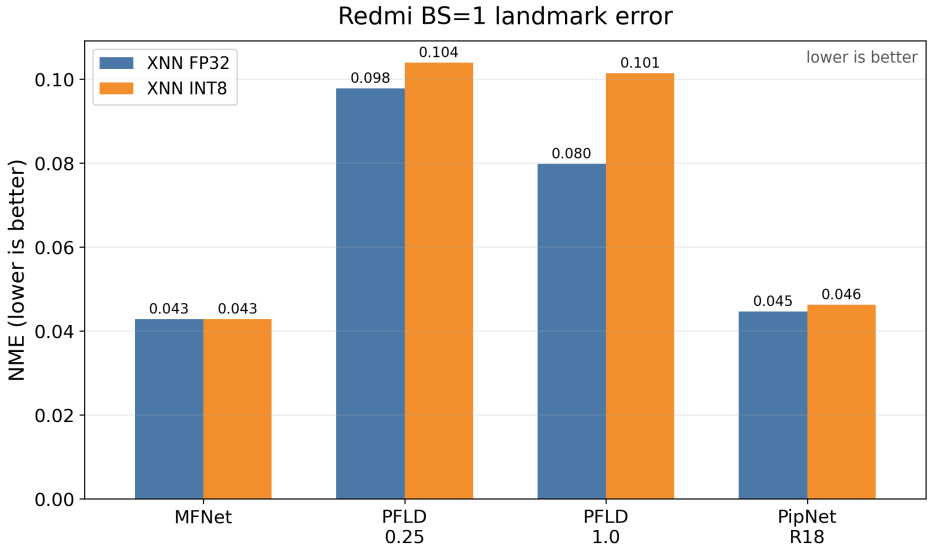


Table 5.1 Focus-device BS=1 XNNPACK metrics. Run FPS is measured benchmark-run throughput, LM inf. means landmark inference, E2E means end-to-end, and ms means milliseconds. Bold values mark the best value in each numeric column: highest run FPS and lowest latency/error.

Model	Quant.	Run FPS	LM inf. [ms]	E2E [ms]	NME
MobileFaceNet	FP32	3.39	184.33	277.18	**0.0428**
MobileFaceNet	INT8	3.46	178.39	272.20	**0.0428**
PFLD-MNv2-0.25	FP32	8.04	9.99	107.34	0.0978
PFLD-MNv2-0.25	INT8	**8.24**	**7.10**	**104.81**	0.1039
PFLD-MNv2-1.0	FP32	5.82	61.54	155.65	0.0798
PFLD-MNv2-1.0	INT8	7.22	23.78	121.56	0.1014
PipNet-R18	FP32	2.28	351.97	421.52	0.0446
PipNet-R18	INT8	5.53	97.46	164.59	0.0463

5.5 Focus-device batch scaling

Figure 5.8 Focus device batch scaling for PFLD-MNv2-1.0 using XNNPACK variants.

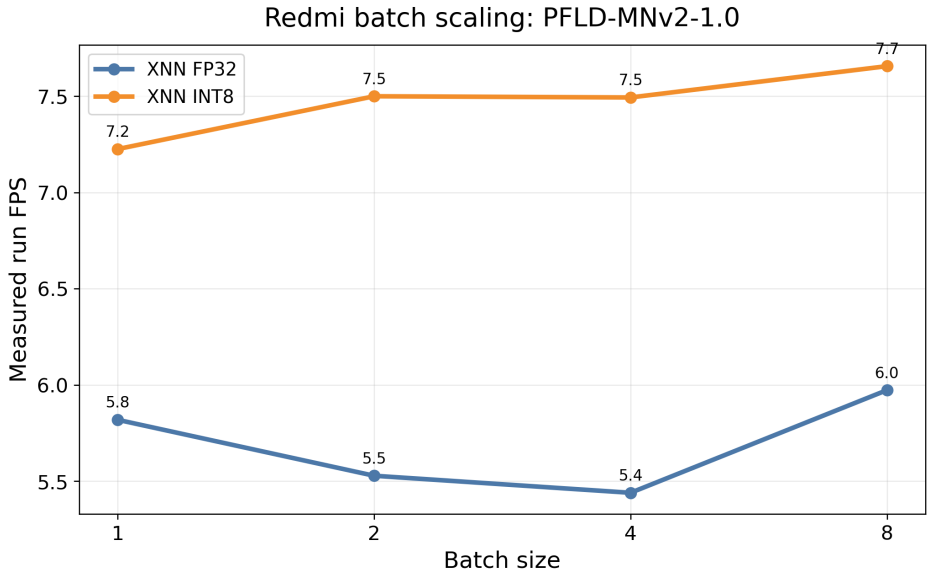


Table 5.2 Focus-device batch scaling in measured run FPS. BS means batch size. Bold marks the highest run FPS in each batch-size column.

Model	Quant.	Run FPS BS=1	Run FPS BS=2	Run FPS BS=4	Run FPS BS=8
PFLD-MNv2-1.0	FP32	5.82	5.53	5.44	5.97
PFLD-MNv2-1.0	INT8	**7.22**	**7.50**	**7.49**	**7.66**
PipNet-R18	FP32	2.28	2.22	2.31	2.27
PipNet-R18	INT8	5.53	5.34	5.52	5.66

Measured run FPS for PFLD-MNv2-1.0 stays in a tight 5.4–6.0 range (FP32) and 7.2–7.7 range (INT8) from BS=1 to BS=8, while PipNet-R18 hovers at ∼2.3 run FPS (FP32) and ∼5.4–5.7 run FPS (INT8). In contrast, per-image landmark inference latency falls almost linearly with batch size: PFLD-MNv2-1.0 INT8 drops from 23.78 ms to 5.29 ms (4.5× across 8× batch), and PipNet-R18 INT8 from 97.46 ms to 27.26 ms (3.6×).

Figures **??** and **??** compare batch scaling across all three devices. A larger batch gives the landmark model more face crops at once, which can improve hardware utilization. This is useful when several faces are present in one frame or when an offline process can group multiple images together. However, it is less useful for a single-face real-time camera stream, because the detector still

Figure 5.9 Focus device batch scaling for PipNet ResNet18 using XNNPACK variants.

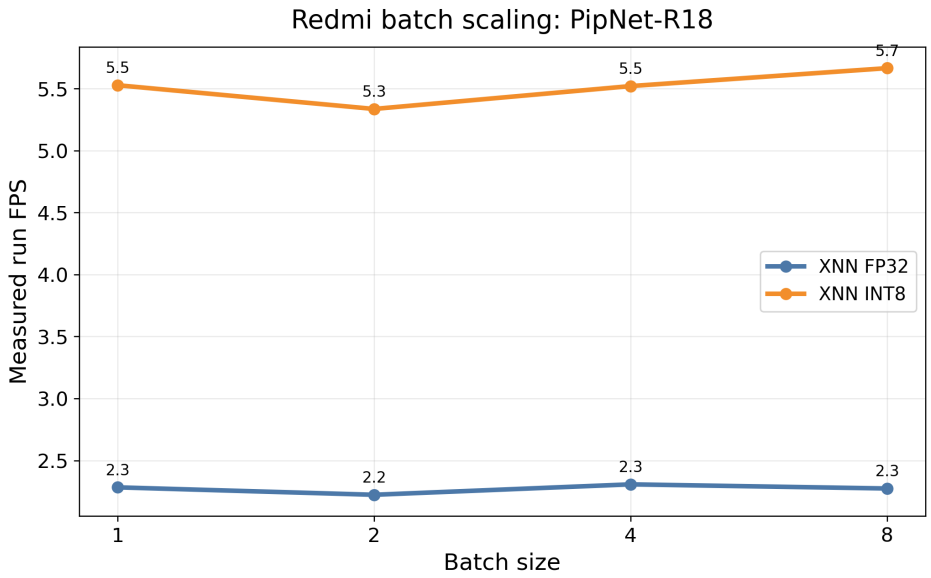


Table 5.3 Focus-device batch scaling in landmark-inference latency. LM means landmark inference and BS means batch size. Bold marks the lowest latency in each batch-size column.

Model	Quant.	LM [ms] BS=1	LM [ms] BS=2	LM [ms] BS=4	LM [ms] BS=8
PFLD-MNv2-1.0	FP32	61.54	37.19	24.78	16.03
PFLD-MNv2-1.0	INT8	**23.78**	**11.81**	**7.37**	**5.29**
PipNet-R18	FP32	351.97	198.58	132.34	103.89
PipNet-R18	INT8	97.46	55.81	37.10	27.26

Figure 5.10 Three-device INT8 XNNPACK batch scaling for PFLD-MNv2-1.0. The horizontal axis is the number of face crops processed together by the landmark model; higher measured run FPS is better.

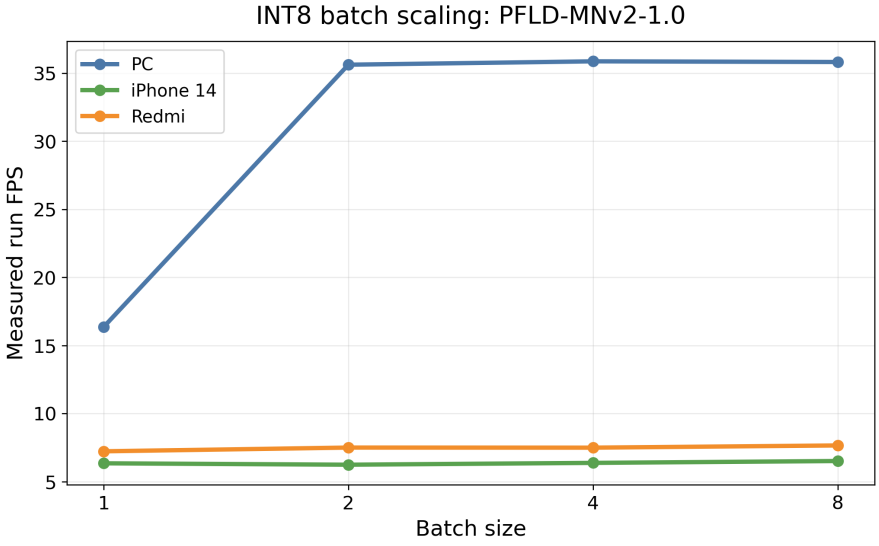
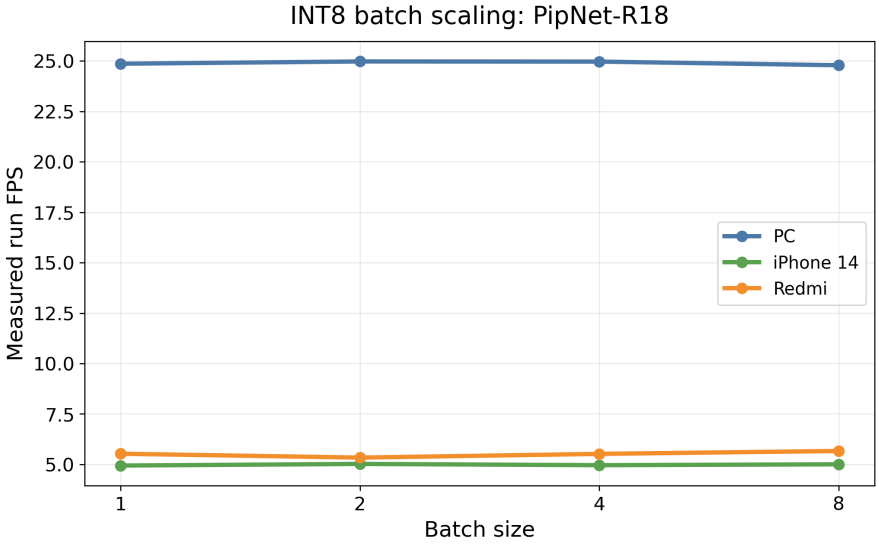


Figure 5.11 Three-device INT8 XNNPACK batch scaling for PipNet-R18. The chart shows whether larger landmark batches improve measured run throughput on PC, iPhone 14, and Redmi.



runs on incoming frames and the application cannot wait too long just to fill a batch.

The PFLD-MNv2-1.0 chart shows that the PC benefits strongly from moving beyond BS=1, while both phones remain almost flat. The implication is that batching helps most when the landmark model is a major part of total runtime and when the hardware can exploit the larger tensor workload. The PipNet-R18 chart is flatter on all devices in run FPS even though landmark-only latency improves with batch size. This is an important practical lesson: optimizing the landmark forward pass can be real and measurable, but measured run FPS may barely move if another stage, such as detection or image handling, has become the bottleneck.

5.6 Quantization deltas and cumulative error behavior

Figure 5.12 Focus device: INT8 vs FP32 measured run-throughput comparison at BS=1.

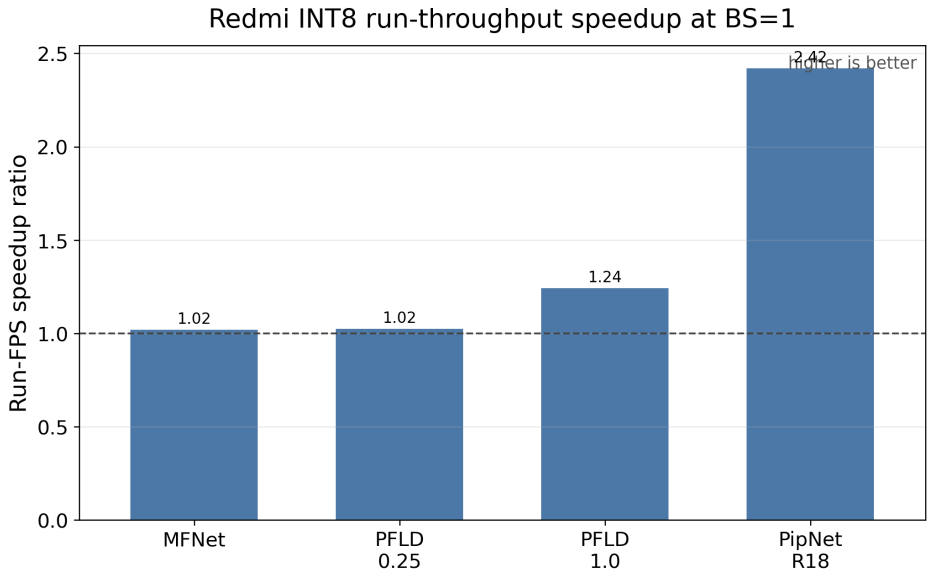


Figure ?? summarises the BS=1 INT8/FP32 run-FPS ratio per model and makes the two-regime story visually explicit. A bar near 1.0× means that INT8 and FP32 are practically the same speed for the measured benchmark run. A bar above 1.0× means INT8 is faster. MobileFaceNet and PFLD-MNv2-0.25 sit almost on the unity line (1.02× and 1.02×), PFLD-MNv2-1.0 reaches 1.24×, and PipNet-R18 is the clear outlier at 2.42×. Therefore, quantization is not a universal speed button; it helps most when the model has enough quantizable computation for the backend to benefit.

Figure 5.13 Focus device: INT8 vs FP32 NME difference at BS=1.

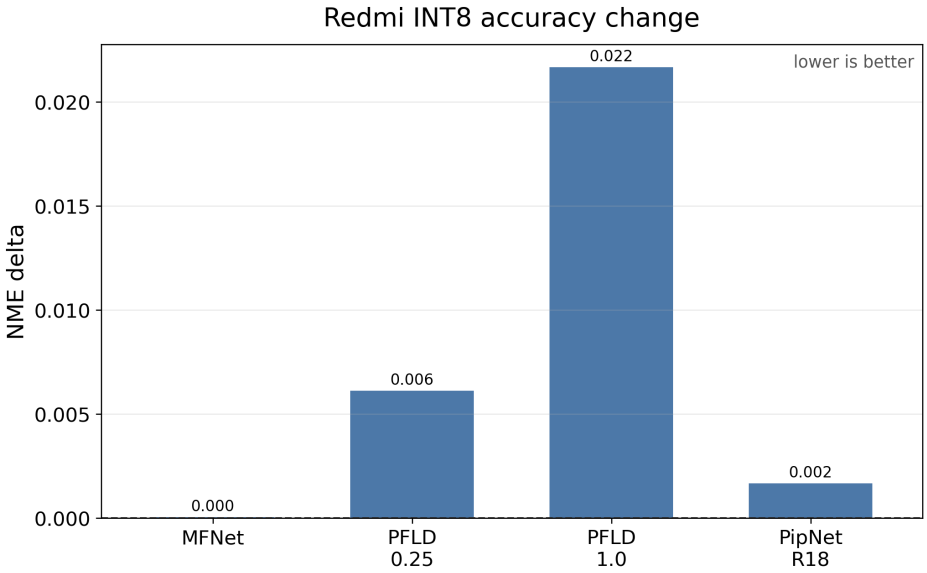


Figure 5.14 Focus device cumulative NME histogram for BS=1 XNNPACK FP32 and INT8 runs. Curves that rise earlier and stay higher indicate that more images have small normalized landmark error.

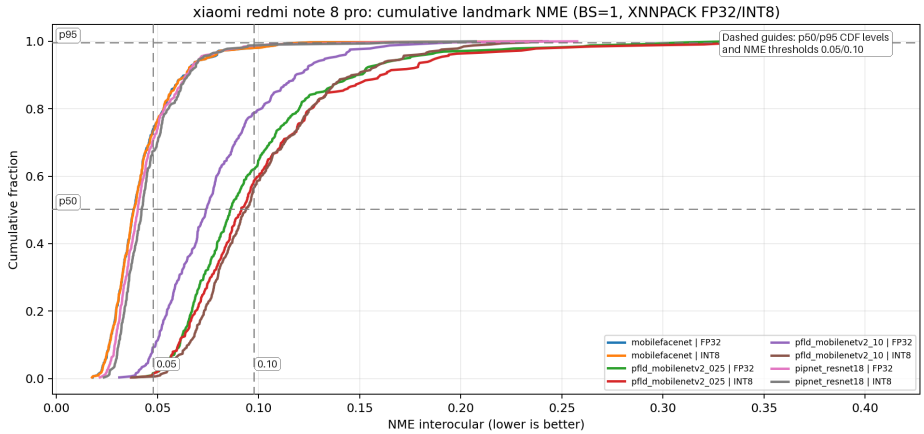


Table 5.4 BS=1 PC INT8 and focus-device INT8 metrics. Bold marks the best value in each column: highest measured run FPS and lowest E2E latency.

Model	PC INT8 run FPS	Focus INT8 run FPS	PC E2E [ms]	Focus E2E [ms]
MobileFaceNet	16.75	3.46	59.64	272.20
PFLD-MNv2-0.25	16.63	**8.24**	59.92	**104.81**
PFLD-MNv2-1.0	16.35	7.22	61.01	121.56
PipNet-R18	**24.86**	5.53	**40.19**	164.59

Figure 5.15 Focus device cumulative RMSE histogram for BS=1 XNNPACK FP32 and INT8 runs. This uses the same cumulative logic as the NME chart, but reports root-mean-square landmark error.

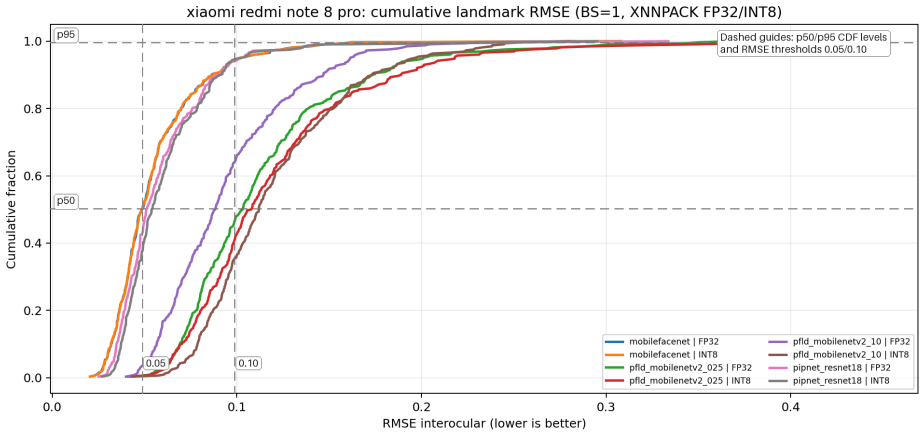


Figure **??** must be read in the opposite direction from the speedup plot. Here, values near zero are good because they mean INT8 changed the error very little. Positive values mean the INT8 model is less accurate than its FP32 version. The cumulative histograms in Figures **??** and **??** provide the deeper view behind those averages. If the FP32 and INT8 curves overlap, quantization does not substantially change the distribution of easy and difficult images. If the INT8 curve shifts to the right or rises more slowly, more images require a larger error threshold before they count as successful. The dashed p50 and p95 lines make this visible without reading exact numbers from every curve: the horizontal crossing near p50 approximates typical-image behavior, while the p95 crossing shows whether the tail cases are still controlled. In these charts, MobileFaceNet and PipNet-R18 preserve similar distributions after quantization, whereas PFLD-MNv2-1.0 separates more visibly. The deployment heuristic is therefore: prefer INT8 for PipNet-R18, treat MobileFaceNet INT8 as a small but mostly free win, and evaluate the PFLD-family accuracy/throughput trade-off against the needs of the actual application.

Table **??** shows the absolute deployment gap. After INT8 quantization on both sides, PC throughput is still 2.0×–4.8× higher than the phone, and the gap is largest for the smallest models (MobileFaceNet at 4.85×).

5.7 iPhone 14 result interpretation

Tables **??** and **??** show the difference between three different backend delegates on iPhone 14. The most striking pattern is that Core ML delivers a remarkably uniform 7.1–8.6 run FPS across all four landmark models—from the smallest PFLD-MNv2-0.25 to the heaviest PipNet-R18—which is only pos-

Figure 5.16 iPhone 14: BS=1 median measured run FPS (device-focused chart pack).

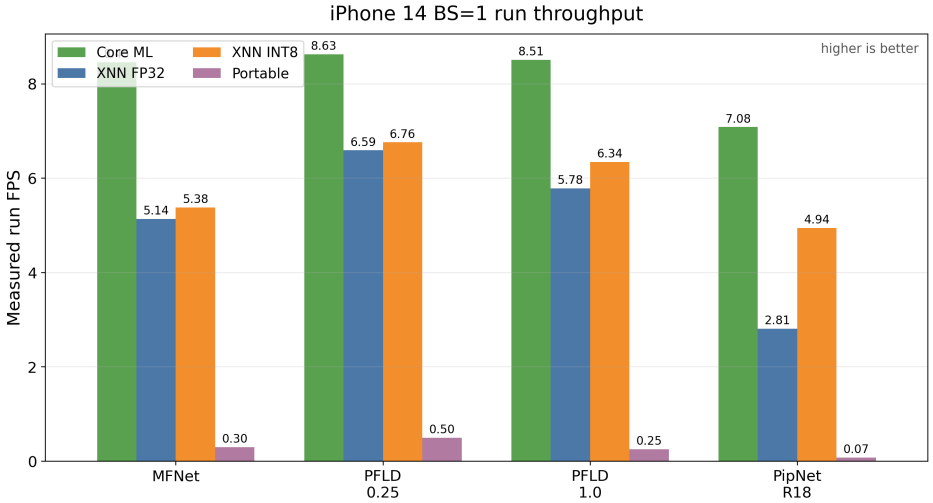


Figure 5.17 iPhone 14: BS=1 median landmark-inference latency.

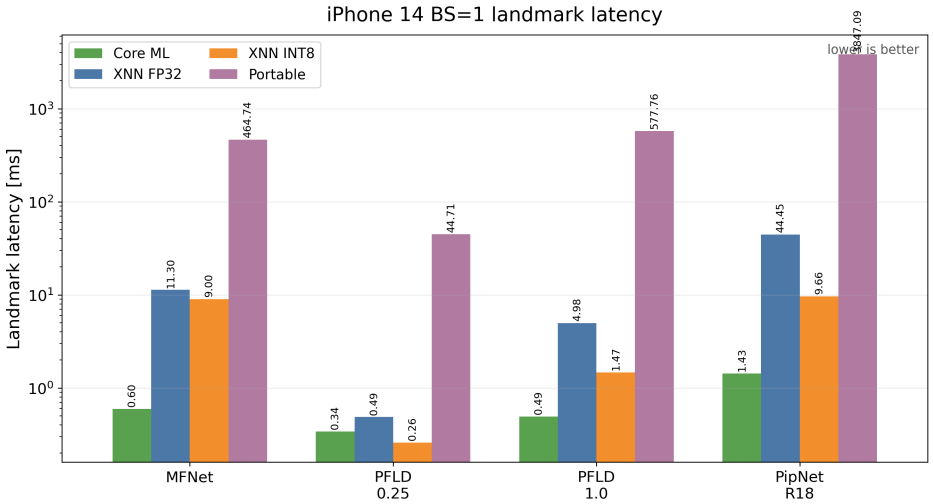


Table 5.5 iPhone 14 BS=1 measured run throughput by backend. XNN means XNNPACK. Bold marks the highest run FPS in each backend column.

Model	Core ML run FPS	XNN FP32 run FPS	XNN INT8 run FPS	Portable run FPS
MobileFaceNet	8.45	5.14	5.38	0.30
PFLD-MNv2-0.25	**8.63**	**6.59**	**6.76**	**0.50**
PFLD-MNv2-1.0	8.51	5.78	6.34	0.25
PipNet-R18	7.08	2.81	4.94	0.07

Figure 5.18 iPhone 14: BS=1 measured run throughput by backend and quantization.

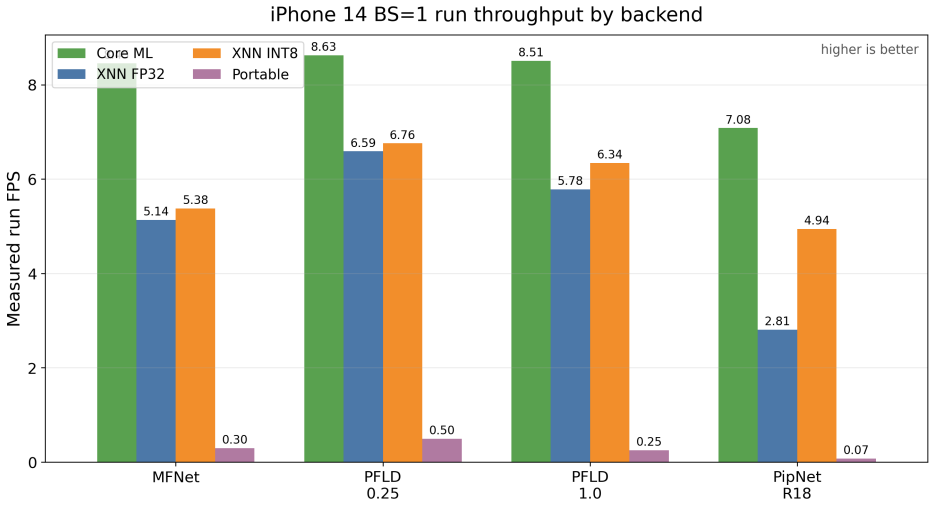


Figure 5.19 iPhone 14: INT8 and FP32 measured run-throughput comparison at BS=1.

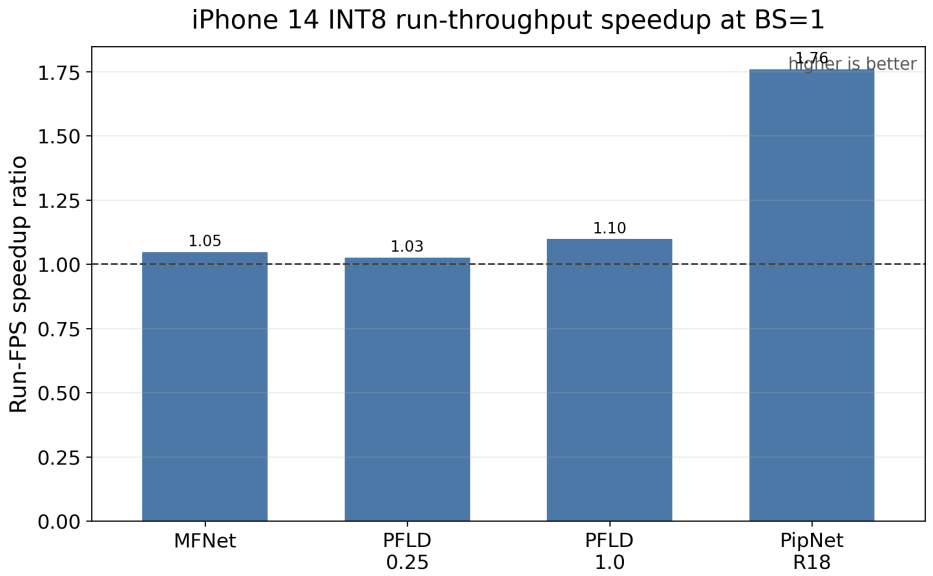


Table 5.6 iPhone 14 BS=1 end-to-end latency by backend. Bold marks the lowest latency in each backend column.

Model	Core ML E2E [ms]	XNN FP32 E2E [ms]	XNN INT8 E2E [ms]	Portable E2E [ms]
MobileFaceNet	**107.35**	177.74	169.33	3374.61
PFLD-MNv2-0.25	107.89	**135.38**	**135.90**	**1999.97**
PFLD-MNv2-1.0	109.89	160.36	141.19	3962.29
PipNet-R18	133.35	338.35	186.01	13698.42

sible because Apple's accelerated kernels reduce the landmark forward pass to essentially zero (median landmark inference of 0.34–1.43 ms for Core ML in the underlying comma-separated values (CSV) file). At that point, the bottleneck moves entirely to the detector (RetinaFace / FaceBoxesV2) and the surrounding pre/post-processing. This is a different operating point from XNNPACK, where landmark inference is still measurable (4–44 ms in FP32) and therefore still tunable through quantization.

Comparing XNNPACK with Portable shows the cost of giving up backend-specific acceleration. Portable is 17–97× slower than Core ML on the same model, ranging from 17.3× for PFLD-MNv2-0.25 to 97.1× for PipNet-R18. Portable is therefore unsuitable for any deployed configuration; it is only useful as a correctness baseline.

The INT8/FP32 split on XNNPACK follows the same model-dependent pattern as on the Xiaomi: PipNet-R18 is again the largest winner (run FPS 2.81 → 4.94, 1.76×; landmark inference 44.5 → 9.7 ms, 4.6×), PFLD-MNv2-1.0 moves modestly (1.10× run FPS, 3.4× landmark inference), and the smaller models barely move ($\leq$ 1.05× run FPS).

Table 5.7 iPhone 14 batch scaling in measured run FPS. BS means batch size. Bold marks the highest run FPS in each batch-size column.

Model	Quant.	Run FPS BS=1	Run FPS BS=2	Run FPS BS=4	Run FPS BS=8
PFLD-MNv2-1.0	FP32	5.78	5.83	5.88	5.91
PFLD-MNv2-1.0	INT8	**6.34**	**6.24**	**6.37**	**6.51**
PipNet-R18	FP32	2.81	2.85	2.85	2.93
PipNet-R18	INT8	4.94	5.02	4.96	5.00

The iPhone 14 batch-scaling pattern is qualitatively the same as on the Xiaomi. PFLD-MNv2-1.0 INT8 lands at 0.17 ms per image at BS=8, and PipNet-R18 INT8 at 1.22 ms; in both cases the landmark forward has effectively been amortised away. Measured run FPS, however, only drifts from 5.78 to 5.91 for PFLD-MNv2-1.0 FP32 and from 2.81 to 2.93 for PipNet-R18 FP32.

Figure 5.20 iPhone 14 batch scaling for PFLD-MNv2-1.0.

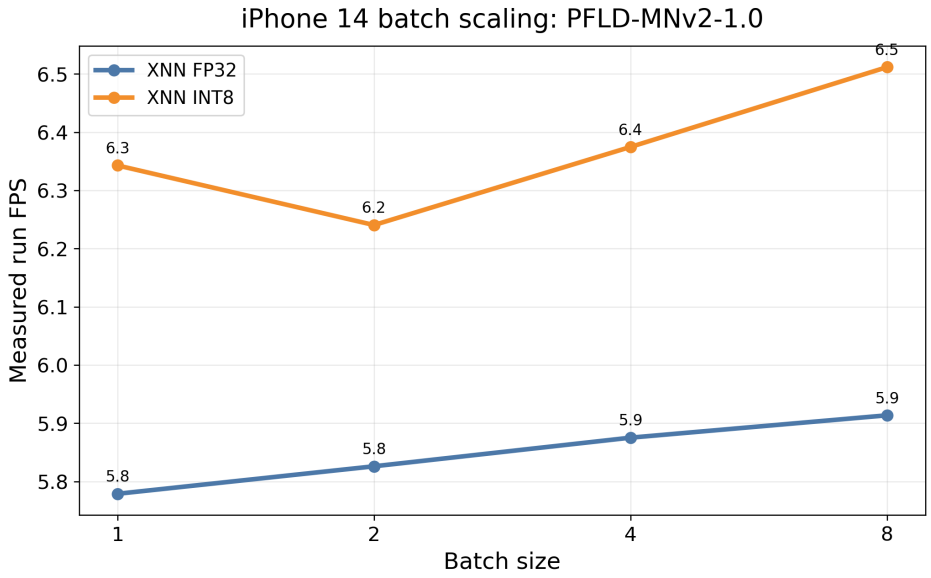


Figure 5.21 iPhone 14 batch scaling for PipNet ResNet18.

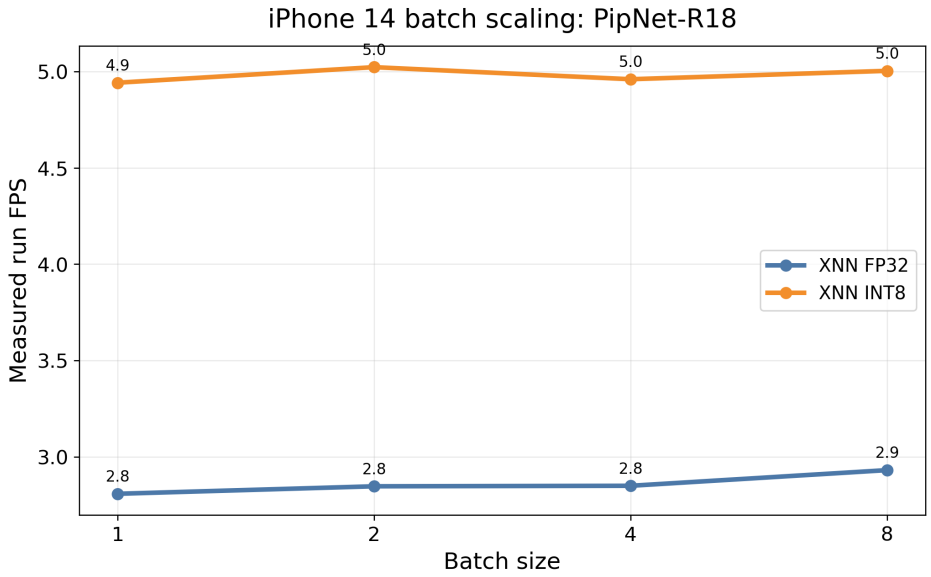


Table 5.8 iPhone 14 batch scaling in landmark-inference latency. LM means landmark inference and BS means batch size. Bold marks the lowest latency in each batch-size column.

Model	Quant.	LM [ms] BS=1	LM [ms] BS=2	LM [ms] BS=4	LM [ms] BS=8
PFLD-MNv2-1.0	FP32	4.98	2.40	1.18	0.59
PFLD-MNv2-1.0	INT8	**1.47**	**0.72**	**0.36**	**0.17**
PipNet-R18	FP32	44.45	22.20	11.26	5.51
PipNet-R18	INT8	9.66	4.82	2.46	1.22

Table 5.9 BS=1 PC INT8 and iPhone 14 INT8 metrics. Bold marks the best value in each column: highest measured run FPS and lowest E2E latency.

Model	PC INT8 run FPS	iPhone INT8 run FPS	PC E2E [ms]	iPhone E2E [ms]
MobileFaceNet	16.75	5.38	59.64	169.33
PFLD-MNv2-0.25	16.63	**6.76**	59.92	**135.90**
PFLD-MNv2-1.0	16.35	6.34	61.01	141.19
PipNet-R18	**24.86**	4.94	**40.19**	186.01

Table **??** sets the deployment context. With the same INT8 XNNPACK configuration on both sides, the PC is 2.46× faster on PFLD-MNv2-0.25, 2.58× on PFLD-MNv2-1.0, 3.12× on MobileFaceNet, and 5.03× on PipNet-R18.

Chapter 6

Conclusion

This thesis presented a full benchmark for landmark detection models on mobile devices with ExecuTorch, from theoretical background and export details to cross platform runtime evaluation.

The thesis delivered:

- implementation and comparison of mobile inference paths on both iOS and Android,
- evaluation of real-time-oriented metrics (ms/frame and run FPS),
- multi-device comparison with reproducible logging and offline analysis,
- quantization study (INT8 PTQ) with speed/accuracy trade-off analysis,
- optional energy-related telemetry as auxiliary data where available.

The results show four main points:

- backend and quantization selection strongly influence runtime;
- INT8 can provide major latency and run-FPS gains for heavier landmark models, but accuracy impact depends on the model;
- increasing batch size improves per-image landmark-inference latency
- PC reference runs are useful for capacity comparison

Overall, the implemented methodology and artifact pipeline provide a reproducible base for future iterations, including model replacement, backend retuning, and extension to more mobile devices.

Appendix A
Reproducibility

This appendix lists the commands used to regenerate the CSV data, charts, and tables referenced in the thesis.

A.1 Combined CSV and chart generation

From the repository root:

```
./.venv/bin/python benchmarks/analysis/plot_runs_dashboard.py \
  --mobile-root benchmarks/results/mobile \
  --pc-root benchmarks/results/pc/runs \
  --focus-device "xiaomi redmi note 8 pro" \
  --max-resolved-batch 8 \
  --output-csv \
    benchmarks/results/combined/combined_runs_from_json.csv \
  --output-dir benchmarks/results/combined/charts
```

A.2 Summary table generation

```
./.venv/bin/python benchmarks/analysis/summarize_benchmarks.py \
  --input benchmarks/results/combined/combined_runs_from_json.csv \
  --output-dir benchmarks/results/combined/tables \
  --focus-device "xiaomi redmi note 8 pro"
```

A.3 INT8 XNNPACK landmark model export with post-training quantization

```
EXPORT_BATCH_SIZES=1 \
EXPORT_DYNAMIC_BATCH=1 \
EXPORT_DYNAMIC_MIN_BATCH=1 \
```

```
EXPORT_DYNAMIC_MAX_BATCH=128 \
XNNPACK_INT8_CALIBRATION_SAMPLES=32 \
XNNPACK_INT8_METHODS=pc,pt \
./models/export_landmark_models_xnnpack_quantized.sh
```

Optional MobileFaceNet stability switch (enabled by default):

```
MOBILEFACENET_STATIC_INT8_DYNAMIC_ACT_WORKAROUND=1
```

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Contents of the attachment

The electronic attachment contains the source code of the benchmark applications, benchmarking scripts, collected run exports and the thesis sources.

- `/`
  - `android/` — Android benchmark app source code
  - `ios/` — iOS benchmark app source code
  - `benchmarks/` — Benchmark harness, analysis scripts, and results
    - `analysis/` — JSON aggregation, charting and table export scripts
    - `results/` — Collected exported runs + generated charts/tables
  - `models/` — Detector and landmark model files used by the apps
  - `thesis/` — LaTeX sources of the thesis
    - `ctufit-thesis.tex` — Main thesis entry point
    - `text/` — Chapter sources